Winsteel Engineering Works Pvt. Ltd.

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SOP STANDARD OPERATING PROCEDURE



SCOPE

- Design & detailing of PEB structures
- Procurement of raw materials
- Fabrication of structural steel members
- Welding & inspection
- Surface preparation & painting
- Assembly & trial fit-up
- Dispatch and supply

APPLICABLE FOR

- Industrial sheds
- Warehouses
- Factory buildings
- Multi-level steel structures
- Commercial PEB structures





SAFETY FIRST
QUALITY ALWAYS

FOLLOW PROCESS
ENSURE EXCELLENCE

TEAMWORK
BUILDS STRENGTH

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2. APPLICABLE STANDARDS

IS 800:2007 – General Construction in Steel
IS 2062 – Structural Steel Specification
IS 875 – Design Loads
IS 1893 – Earthquake Resistant Design
IS 9595 –Gas Metal Arc Welding
IS 816 – Welding Code
AWS D1.1 – Structural Welding Code
SSPC / ISO 8501 – Surface Preparation Standards

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3. RESPONSIBILITIES

Department	Responsibility
Design Department	Structural analysis & detailing
QA/QC Department	Inspection & quality control
Production Department	Fabrication & assembly
Purchase Department	Procurement of approved materials
Dispatch Department	Packing & delivery

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PROCESS FLOW CHART



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5. DETAILED PROCEDURE

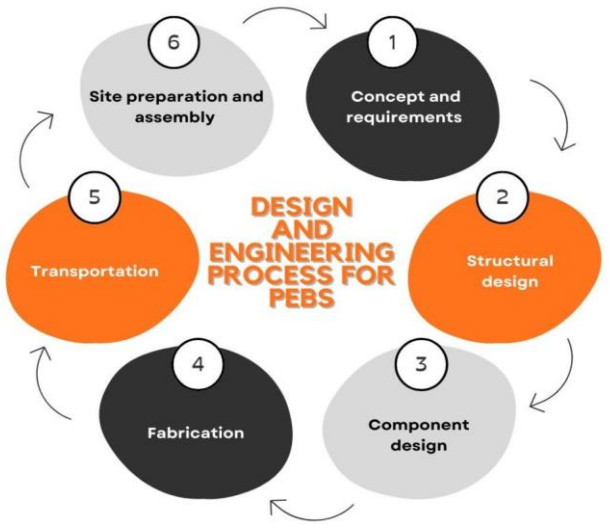
5.1 Design & Engineering

Procedure

- Review client drawings and specifications.
- Perform structural analysis using approved software.
- Prepare:
  - GA Drawings
  - Fabrication Drawings
  - Connection Details
  - BOM (Bill of Materials)
- Ensure design compliance with IS codes.

Deliverables

- Approved drawings
- Design calculations
- Material list



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5.2 Raw Material Procurement

Procedure

- Procure structural steel from approved vendors.
- Verify:
  - Mill Test Certificates (MTC)
  - Grade & thickness
  - Heat number traceability

Inspection

- Visual inspection
- Dimension verification
- Material traceability marking

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5.3 Cutting & Fabrication

Procedure

- CNC/Plasma/Laser/ oxy-fuel/ Profile cutting as per drawings.
- Mark identification numbers on members.
- Prepare built-up sections and assemblies.

Equipment Used

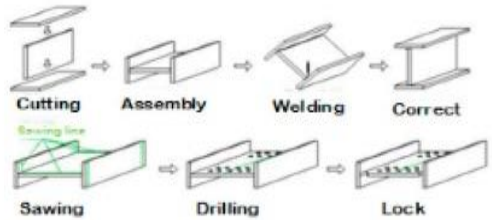
- CNC Cutting Machine
- Band Saw
- Radial Drill
- Plate Bending Machine

Manufacturing Process of Prefabricated Steel Structures

Cutting



Assembly



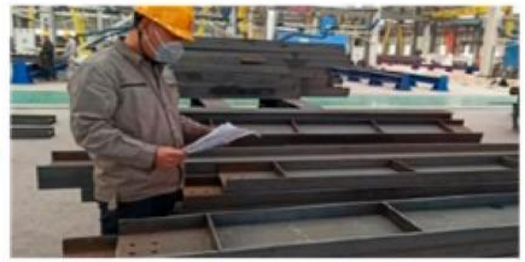
Welding



Painting



Correct



Shipment



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5.4 Fit-Up & Welding

Procedure

- Align members using approved fixtures.
- Maintain dimensions and tolerances.
- Carry out welding as per approved WPS.

Welding Process

- GMAW / SAW / SMAW

Inspection

- Visual Inspection
- Weld Size Verification
- NDT (if required)



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5.5 Quality Control Inspection

Checks

- Dimensional inspection
- Weld quality inspection
- Alignment and straightness
- Bolt hole matching

Testing

- DPT / UT / MPI (as applicable)

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5.6 Surface Preparation & Painting

Surface Preparation

- Mechanical cleaning / shot blasting
- Remove rust, oil, and scale

Painting Procedure

- Apply primer immediately after cleaning.
- Maintain DFT as per specification.

Typical Coating System

Total DFT: 100–120 Microns Minimum

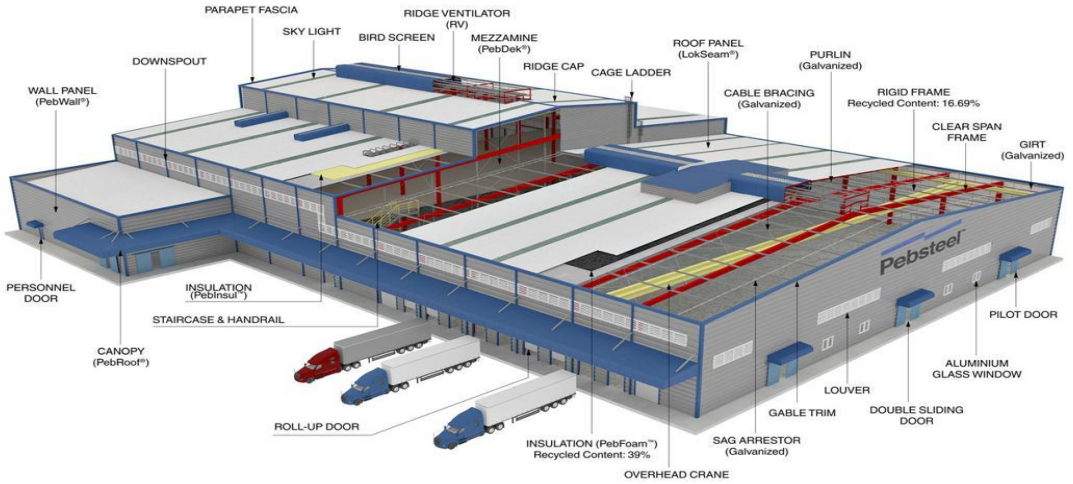
Coat	DFT
Primer	40–60 Microns
Intermediate	40 Microns
Finish Coat	40 Microns

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5.7 Assembly

Procedure

- Carry out trial fit-up of major members.
- Check:
  - Alignment
  - Bolt matching
  - Connection fitment



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5.8 Packing & Dispatch

Procedure

- Mark members clearly.
- Protect painted surfaces.
- Prepare dispatch list and packing list.



Documents

- Inspection reports
- Test certificates
- Delivery challan
- Packing list

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6. SAFETY REQUIREMENTS

- PPE mandatory for all workers.
- Follow welding and lifting safety procedures.
- Use calibrated lifting tools and tackles.
- Fire extinguishers shall be available near welding areas.

7. QUALITY RECORDS

Maintain:

- Material inspection reports
- Welding inspection reports
- Calibration certificates
- Paint DFT reports
- Dispatch records

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8. ACCEPTANCE CRITERIA

Activity	Acceptance Requirement
Material	As per approved grade
Welding	No visible defects
Dimensions	Within drawing tolerance
Painting	Required DFT achieved
Final Inspection	Client/QA approved

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9. CONCLUSION

This Standard Operating Procedure (SOP) establishes a structured and systematic approach for the execution of Pre-Engineered Building (PEB) fabrication activities, ensuring compliance with approved drawings, technical specifications, applicable codes, and client requirements.

The procedure emphasizes effective quality assurance, process control, material traceability, safety compliance, and inspection at every stage of fabrication to achieve consistent workmanship and reliable structural performance.

- **By implementing this SOP, the organization ensures:**
- High standards of fabrication quality and dimensional accuracy
- Safe and efficient manufacturing practices
- Timely project execution and delivery
- Compliance with statutory, contractual, and international standards
- Continuous monitoring, documentation, and accountability throughout the fabrication process

This SOP serves as a guiding framework for maintaining operational excellence, customer satisfaction, and continual improvement in PEB manufacturing and project execution.

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THANK YOU

